

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____

Function Families

Example (Name the parent function):

$f(x) = 2^x$ is the **exponential** parent. It passes through $(0, 1)$, never touches the x-axis, and its domain is all real numbers.

Directions: Name the parent family and give the domain.

1. $f(x) = x^2$

4. $f(x) = \sqrt{x}$

2. $f(x) = |x|$

5. $f(x) = x^3$

3. $f(x) = \frac{1}{x}$

Challenge Problems

6. $f(x) = \log_2 x$ — name the family and give the domain.

Answer Key and Procedure

Procedure

1. Read the rule and match it to a known family.
2. Recall the family's basic shape and key point.
3. State the domain (watch for square roots and denominators).

Answers

1. Quadratic; domain all real numbers
2. Absolute value; domain all real numbers
3. Rational (reciprocal); domain $x \neq 0$
4. Square root; domain $x \geq 0$
5. Cubic; domain all real numbers
6. Logarithmic; domain $x > 0$